

Using graphs for clustering and structure extraction

Generalisation to higher-order interactions

PhD defence, Tuesday 8 September 2026

Learning Centre SophiaTech, Université Côte d'Azur

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ÉCOLE UNIVERSITAIRE DE RECHERCHE
**SYSTÈMES NUMÉRIQUES
POUR L'HUMAIN**

01

Prologue: an old riddle





An old riddle...

Un classificateur non-supervisé utilisant les
complexes simpliciaux
(avec une application à la *stylométrie*).

LOUIS HAUSEUX
Encadré par BARTŁOMIEJ BŁASZCZYSZYN

juillet 2017

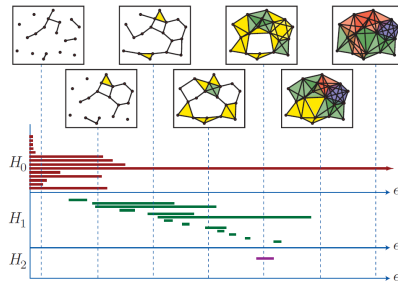
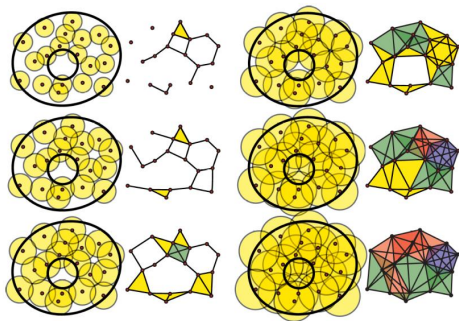
Avant-propos

Internship at Inria Paris, DYOGENE-MATHNET team [\[HAL 17\]](#)

[\[HAL 17\]](#) L. Hauseux (supervised by B. Błaszczyszyn), "Un classificateur non supervisé utilisant les complexes simpliciaux", Inria Paris, 2017. HAL: hal-01597846.



The fashionable topic: *persistent* homology



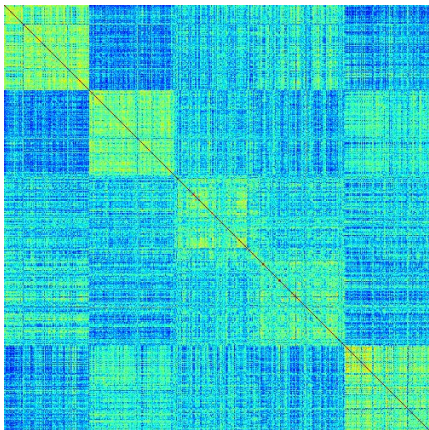
Grow a radius r ; watch structures be **born, live, die** [Ghrist 08] [ToMATo 13]

[Ghrist 08] R. Ghrist, "Barcodes: the persistent topology of data", Bulletin of the AMS, 2008.

[ToMATo 13] F. Chazal, L. J. Guibas, S. Y. Oudot, P. Skraba, "Persistence-based clustering in Riemannian manifolds" (ToMATo), *J. ACM*, 60(6), art. 41, 2013.



...with an application to *stylometry*

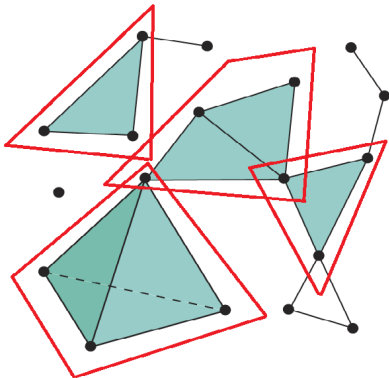


- ▶ A **Gram** matrix: 500×500 similarity between newspaper articles [HAL 17]
- ▶ **5 authors** (anonymous), 100 articles each (style statistics: vocabulary, parts of speech, punctuation, *etc.*)
- ▶ **Goal (unsupervised)**: give each author back their 100 articles

Here the articles are *sorted* by author; the algorithm only sees a shuffled matrix



A heuristic inspired by persistence

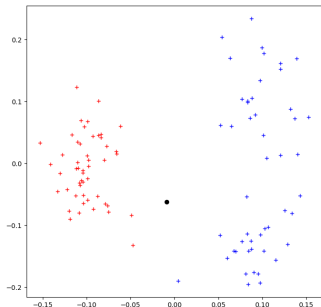


- ▶ Grow the similarity threshold
- ▶ Watch the “**triangle-connected**” components (k -simplices glued along facets)
- ▶ See them be **born, grow, merge**
- ▶ Stop: a proportion P is covered

The “polyhedra” returned ($k = 2$) [HAL 17]



And... it works well



Principal component analysis: two authors out of five

- ▶ **2 authors:** 1 error, 1 unclassified article (out of 100)
- ▶ Better than the *k*-means algorithm...
- ▶ ... and better than *semi-supervised* Gaussian mixtures
- ▶ And with **5 authors:** 4 clusters recovered

Why does it work so well?



Outline

- I) Prologue: an old riddle
- II) From *Single-Linkage* to its foundations
- III) Going **higher-order**: the exact K -NN hierarchy
- IV) **Percolation** as a performance measure
- V) Applications: 3D and 4D LiDAR, training-free
- VI) Towards more structured data
- VII) Perspectives: the **hierarchical** path
- VIII) Conclusion

02

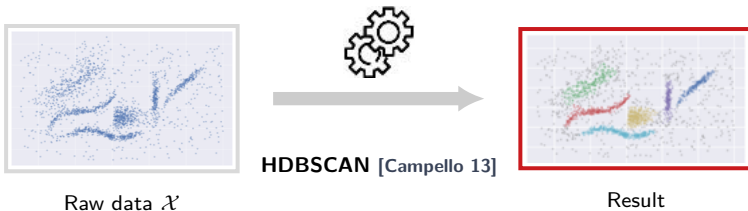
From *Single-Linkage* to its foundations





Clustering on Euclidean data $\mathcal{X} \subset \mathbb{R}^p$

The Hartigan model [Hartigan 81]: high-density clusters

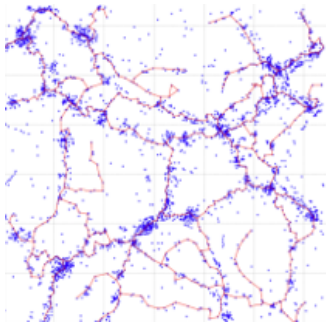


[Hartigan 81] J. A. Hartigan, "Consistency of Single Linkage for high-density clusters", JASA, 1981.

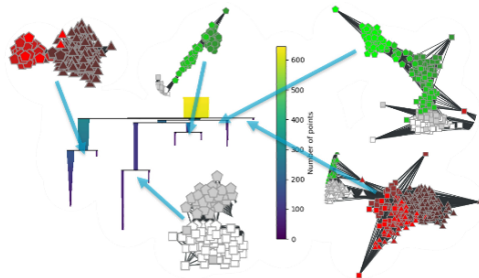
[Campello 13] R. Campello, D. Moulavi, J. Sander, "Density-based clustering based on hierarchical density estimates" (HDBSCAN), PAKDD, 2013.



Two examples from the thesis



3D cosmology: cosmic filaments [CN 23]



Olive oil ($\mathbb{R}^8$): resulting hierarchy [EUSIPCO 24]
[ANS 26]

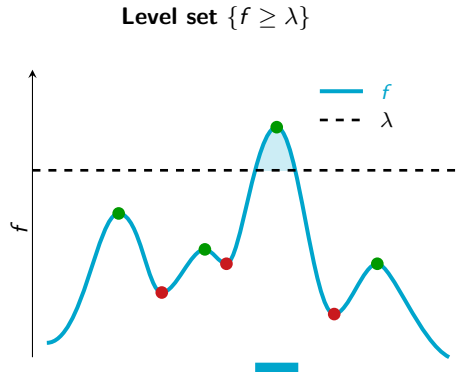
[CN 23] L. Hauseux, K. Avrachenkov, J. Zerubia, "Graph based approach for galaxy filament extraction", *Complex Networks & Their Applications*, Menton, 2023.

[EUSIPCO 24] L. Hauseux, K. Avrachenkov, J. Zerubia, "Benefits of hypergraphs for density-based clustering", *EUSIPCO*, Lyon, 2024.

[ANS 26] L. Hauseux, K. Avrachenkov, J. Zerubia, "Generalization of Single-Linkage with higher-order interactions", *Applied Network Science*, 11(9), 2026.



The Hartigan model: the cluster hierarchy

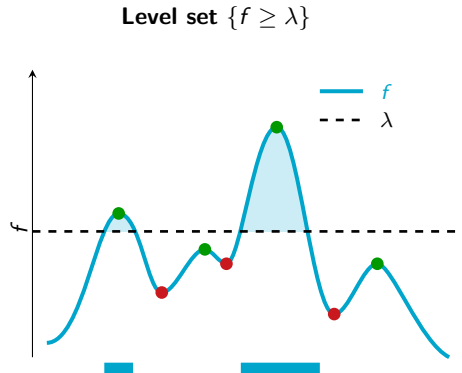


Hierarchical tree

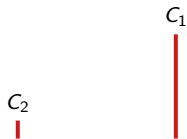
C_1



The Hartigan model: the cluster hierarchy

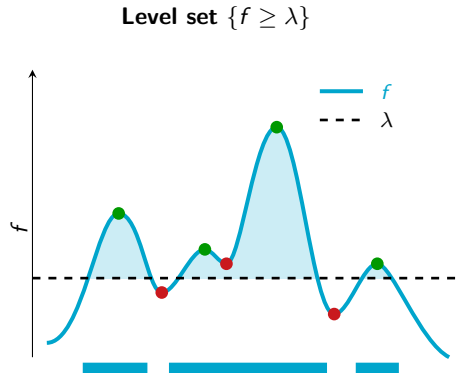


Hierarchical tree

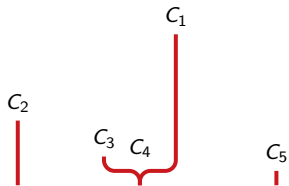




The Hartigan model: the cluster hierarchy



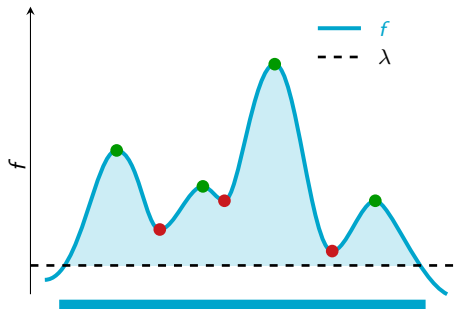
Hierarchical tree



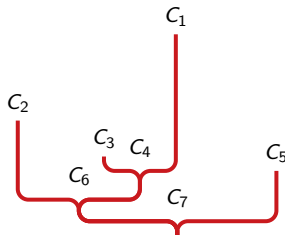


The Hartigan model: the cluster hierarchy

Level set $\{f \geq \lambda\}$



Hierarchical tree





In practice: estimating the density

$$\hat{f} \leftarrow \hat{f}_{K\text{-NN}}$$

K -nearest neighbour estimator [Biau–Devroye 15]

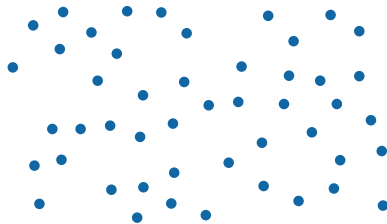
$$\forall y \in \mathbb{R}^p, \quad \hat{f}_{K\text{-NN}}(y) = \frac{K}{n \omega_p r_K(y)^p}$$

$$r_K(y) = \min \{r \geq 0 \mid |B(y, r) \cap (\mathcal{X} \setminus \{y\})| \geq K\}$$

- ▶ y : a point of the **space**, in the point cloud or not
- ▶ n : number of points in $\mathcal{X}$
- ▶ ω_p : volume of the unit ball of $\mathbb{R}^p$
- ▶ $r_K(y)$: distance from y to its K th nearest neighbour; if $y \in \mathcal{X}$, y itself is excluded

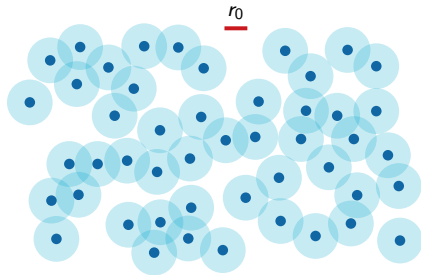


The case $K = 1$: a geometric “miracle”





The case $K = 1$: a geometric “miracle”

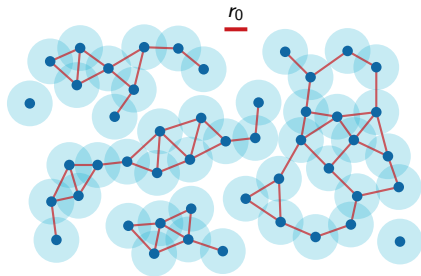


$$\begin{aligned}\hat{\lambda}(y) &\geq \lambda_0 \\ \Leftrightarrow r_1(y) &\leq r_0\end{aligned}$$

Level sets
= **Boolean model** [Meester–Roy 96]



The case $K = 1$: a geometric “miracle”

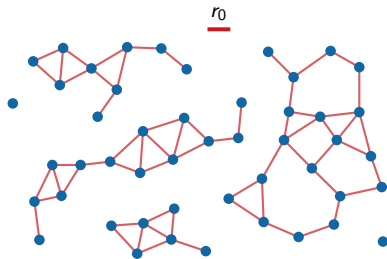


$$\hat{\lambda}(y) \geq \lambda_0$$
$$\Leftrightarrow r_1(y) \leq r_0$$

High-density clusters
= **connected components**
of a **geometric graph** [Penrose 03]



The case $K = 1$: a geometric “miracle”

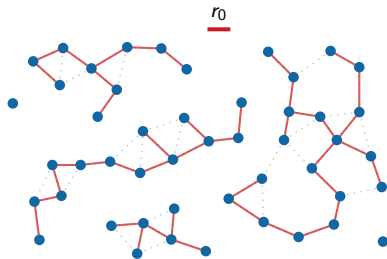


$$r_1(y) \leq r_0$$

High-density clusters
= **connected components**
of a **geometric graph** [Penrose 03]



The case $K = 1$: a geometric “miracle”



$$r_1(y) \leq r_0$$

High-density clusters
= **connected components**
of the pruned **minimum spanning tree**

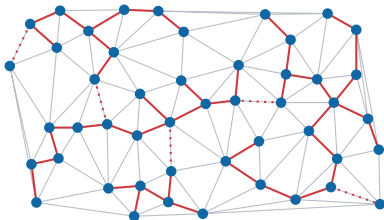


The minimum spanning tree and the Delaunay triangulation

This hierarchy has a name:

Single-Linkage [Florek 51]

[Gower–Ross 69]



The Euclidean minimum spanning tree (MST) is a **subgraph** of GABRIEL, itself a subgraph of DELAUNAY

[Boissonnat 18]:

minimum spanning tree
 $\subseteq$ GABRIEL $\subseteq$ DELAUNAY

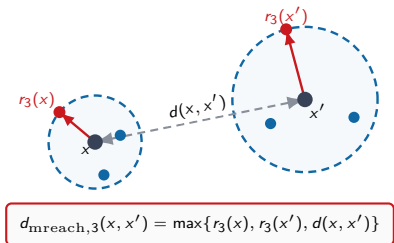
In the plane $\mathbb{R}^2$: $\mathcal{O}(n \log n)$

[Florek 51] K. Florek, J. Łukaszewicz, J. Perkal, H. Steinhaus, S. Zubrzycki, "Sur la liaison et la division des points d'un ensemble fini", Colloq. Math., 1951.

[Gower–Ross 69] J. C. Gower, G. J. S. Ross, "Minimum spanning trees and Single-Linkage cluster analysis", JRSS C, 1969.

[Boissonnat 18] J.-D. Boissonnat, F. Chazal, M. Yvinec, *Geometric and Topological Inference*, Cambridge University Press, 2018.

The chaining effect and *Robust Single-Linkage*

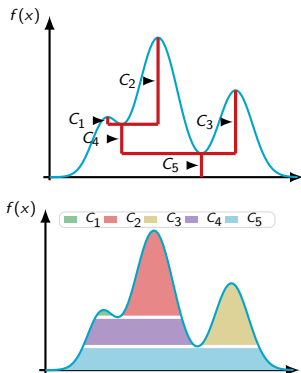


Chaining effect: a chain of points is enough to merge two clusters

Robust Single-Linkage [Chaudhuri–Dasgupta 10]:

- ▶ The *core distance* $r_K(x)$ estimates the local density
- ▶ The distance becomes the *mutual reachability*
- ▶ Isolated points are pushed away

The descendants: DBSCAN, then HDBSCAN



DBSCAN [Ester 96]: *core points* ($r_K(x) \leq r$) vs. *border points*: brings back the cluster boundary

HDBSCAN [Campello 13]: varies the threshold r continuously and extracts clusters by **excess of mass**

$$E(C) = \int_C (f(x) - \lambda) dx$$

[Ester 96] M. Ester, H.-P. Kriegel, J. Sander, X. Xu, "A density-based algorithm for discovering clusters" (DBSCAN), KDD, 1996.

[Campello 13] R. Campello, D. Moulavi, J. Sander, "Density-based clustering based on hierarchical density estimates" (HDBSCAN), PAKDD, 2013.

03

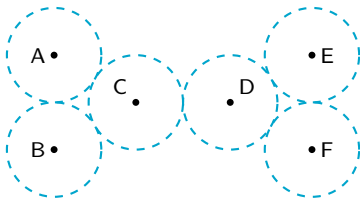
Going **higher-order**: the exact K -NN hierarchy



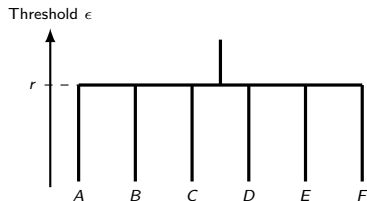


What do Robust SL and (H)DBSCAN do for $K = 2$?

Point cloud and balls of radius r



Resulting hierarchical tree

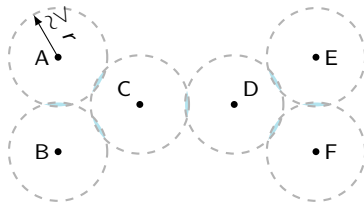


A **flat** dendrogram: all the fine hierarchical information is lost

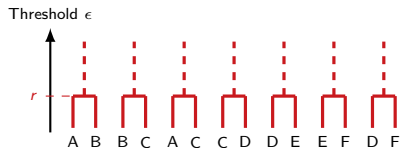


The true 2-NN hierarchy: scale r

The 2-intersections of balls at level r



Hierarchical tree

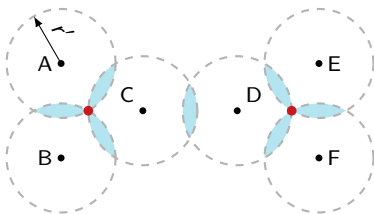


Step 1: the 7 segments merge at threshold r



The true 2-NN hierarchy: scale r'

Triangles and 3-intersections at r'



Hierarchical tree

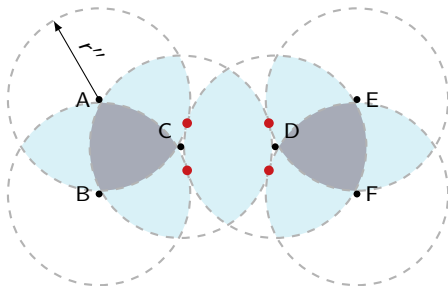


Step 2: the triangles merge at r' via the 3-intersections. Edge CD stays isolated

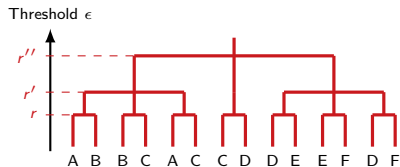


The true 2-NN hierarchy: scale r''

Global connectivity at r''



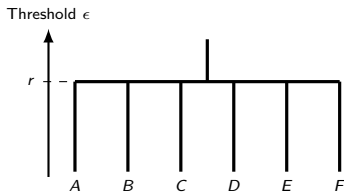
Hierarchical tree



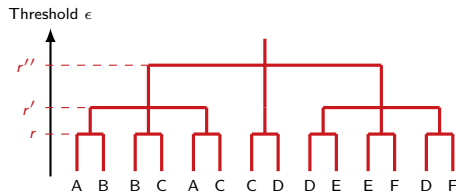
Step 3: global connectivity at r'' through the tangency points

The diagnosis

Robust SL ($K = 2$)



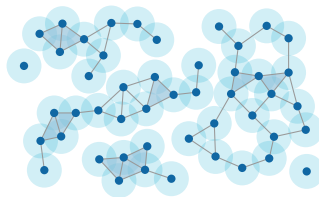
True 2-NN hierarchy



An asymmetry: order- K condition on the **points**, but order-1 connectivity through the **edges**

The fix: the Čech complex and the K -polyhedra

- ▶ **Geometric graph** $\longrightarrow$ **Čech complex**:
a K -simplex $\iff K + 1$ balls with non-empty intersection
- ▶ **Connected components** $\longrightarrow$ **K -polyhedra**:
($K - 1$)-simplices glued along K -simplices
- ▶ For $K \geq 2$, clusters may **overlap**



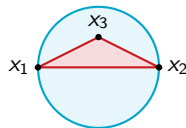
Theorem [ANS 26] [EUSIPCO 24]: at every level, the K -polyhedra identify *exactly* the discrete high-density clusters of the estimator $\hat{f}_{K\text{-NN}}$. For $K = 1$: *Single-Linkage*

[ANS 26] L. Hauseux, K. Avrachenkov, J. Zerubia, "Generalization of Single-Linkage with higher-order interactions", *Applied Network Science*, 11(9), 2026.
[EUSIPCO 24] L. Hauseux, K. Avrachenkov, J. Zerubia, "Benefits of hypergraphs for density-based clustering", *EUSIPCO*, Lyon, 2024.

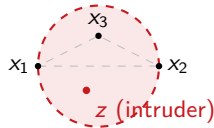
What causes the merges? The Gabriel condition

Edge of the minimum spanning tree $\rightarrow$ K -splitting simplex

- ▶ **Definition [ANS 26]:** a simplex is said to be **K -splitting** if it causes a K -NN merge
- ▶ **Case $K = 1$:** the 1-splitting simplices are exactly the edges of the MST
- ▶ **Definition [ANS 26]:** a K -simplex is said to be **Gabriel** if its miniball contains no other point
- ▶ **Theorem [ANS 26]:** every K -splitting simplex is GABRIEL



Gabriel triangle (empty ball)



Non-Gabriel (non-empty ball)



K -minimum spanning tree and order- K Delaunay mosaic

- ▶ **Critical merges:** the K -splitting simplices, all K -GABRIEL (empty miniball)
- ▶ **Minimum spanning tree** $\rightarrow$ K -MST:
pruned at r , it returns the K -polyhedra [ANS 26]
- ▶ **Delaunay** $\rightarrow$ **order- K mosaic:**
it carries every K -GABRIEL simplex [Edelsbrunner–Osang 23]
- ▶ For $K = 2$: everything lies in DELAUNAY, in $\mathcal{O}(n \log n)$ in the plane

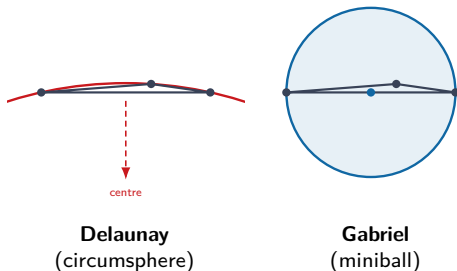
Order 1 vs. order K analogy

	$K = 1$	$K \geq 2$
Vertices	points	$(K - 1)$ -simplices
Merges	edges	K -simplices
Condition	GABRIEL edge	K -GABRIEL
Domain	DELAUNAY	order- K DELAUNAY

[Edelsbrunner–Osang 23] H. Edelsbrunner, G. Osang, “A simple algorithm for higher-order Delaunay mosaics and alpha shapes”, *Algorithmica*, 2023.
[ANS 26] L. Hauseux, K. Avrachenkov, J. Zerubia, “Generalization of Single-Linkage with higher-order interactions”, *Applied Network Science*, 11(9), 2026.



Why Gabriel rather than the higher-order Delaunay mosaic?



- ▶ Near-degenerate, the circumsphere admits **arbitrarily distant** witnesses; those of the miniball **can be localised**
- ▶ A **well-separated pair decomposition** [Callahan–Kosaraju 95] then rules out **blocks** of candidates in parallel; DELAUNAY stays sequential

[Callahan–Kosaraju 95] P. B. Callahan, S. R. Kosaraju, "A decomposition of multidimensional point sets with applications to k -nearest-neighbors and n -body potential fields", *J. ACM*, 42(1), 1995.

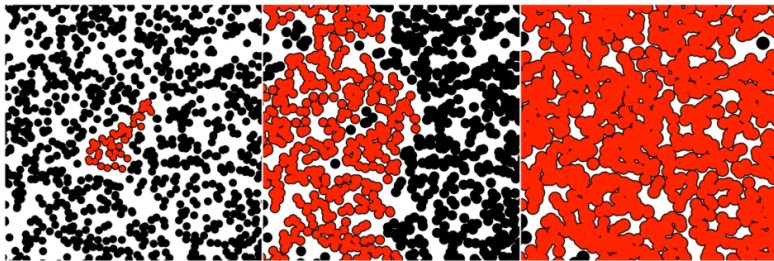
04

Percolation as a performance measure





The phenomenon at the heart of it all: continuum percolation



$$r < r_c$$

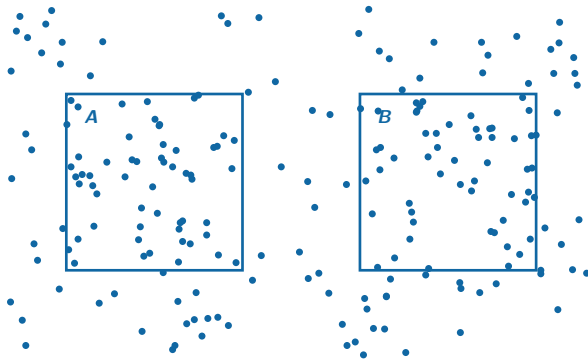
$$r = r_c$$

$$r > r_c$$

Above a critical threshold, a **giant component** (in red) emerges [Penrose 03]



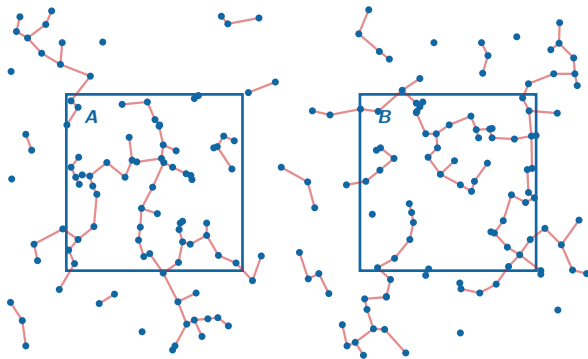
Why *Single-Linkage* fails in dimension ≥ 2



Two dense clusters, A and B (the squares), on a low-density background



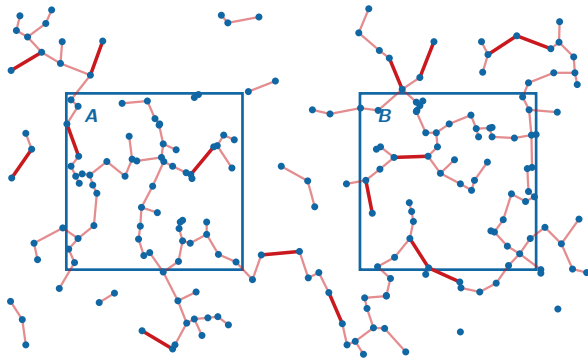
Why *Single-Linkage* fails in dimension ≥ 2



Minimum spanning tree pruned at radius r_1 : percolation in each square, not in the background



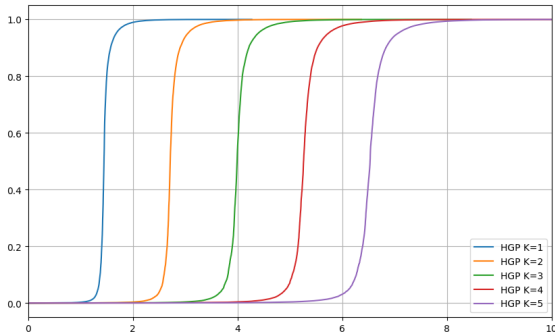
Why *Single-Linkage* fails in dimension ≥ 2



At radius $r_2 > r_1$, **the background percolates too**: *A and B merge before they are fully recovered*



The recoverable fraction

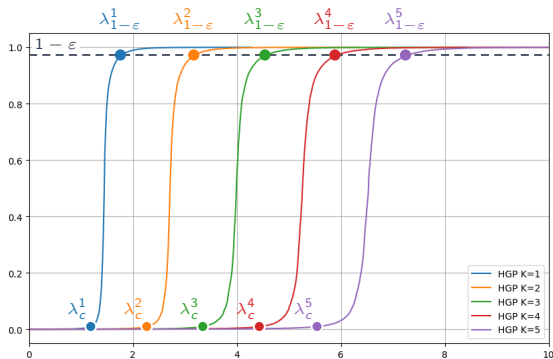


Fraction of points in the giant component [ANS 26]

- ▶ On the x-axis: the intensity λ ; on the y-axis: the fraction of points in the **giant component**
- ▶ Below the critical threshold λ_c : nothing
- ▶ A cluster's **recoverable fraction** is read off the curve



Quantifying it: the *percolation rate*



$$v = \frac{\lambda_c}{\lambda_{1-\epsilon}}$$

► $\lambda_{1-\epsilon}$: the cluster is almost **whole**

► The **closer v is to 1**,
the better the algorithm

[SIGMETRICS SRC 24] [CN 24]

Percolation probability of the K -polyhedra [ANS 26]

[SIGMETRICS SRC 24] L. Hauseux, "How can we theoretically measure the performance of density-based clustering algorithms?", ACM SIGMETRICS SRC, Venice, 2024 (second prize).

[CN 24] L. Hauseux, K. Avrachenkov, J. Zerubia, "Hypergraphs, percolation and hierarchical clustering", *Complex Networks & Their Applications*, Istanbul, 2024.

[ANS 26] L. Hauseux, K. Avrachenkov, J. Zerubia, "Generalization of Single-Linkage with higher-order interactions", *Applied Network Science*, 11(9), 2026.



Measured percolation rates

K	$p = 2$			$p = 3$			$p = 4$		
	HGP	Robust SL	DBSCAN	HGP	Robust SL	DBSCAN	HGP	Robust SL	DBSCAN
1	0.735	0.735	0.735	0.563	0.563	0.563	0.362	0.362	0.362
2	0.779	0.689	0.735	0.646	0.462	0.563	0.451	0.271	0.362
3	0.800	0.632	0.753	0.690	0.412	0.582	0.497	0.247	0.378
4	0.817	0.598	0.767	0.714	0.400	0.605	0.500	0.246	0.409
5	0.826	0.584	0.779	0.732	0.399	0.625	0.534	0.236	0.415

Percolation rate measurements [ANS 26]

- ▶ **K -polyhedra**: the rate **increases** with K , in dimension 2, 3 and 4
- ▶ **Robust SL**: it **decreases**; pruning the boundary throws away part of the cluster
- ▶ **DBSCAN**: it brings back that pruned boundary, but its cores percolate too early

[ANS 26] L. Hauseux, K. Avrachenkov, J. Zerubia, "Generalization of Single-Linkage with higher-order interactions", *Applied Network Science*, 11(9), 2026.

05

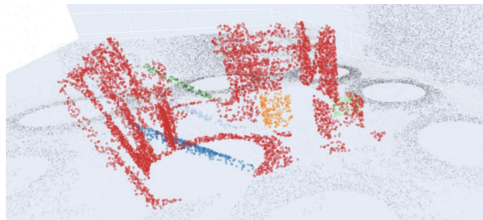
Applications: 3D and 4D LiDAR, training-free





The hierarchy becomes a search space

- ▶ The hierarchical tree is not just an output: we can **inject geometric priors** into it
- ▶ **3D anomaly detection**: comparison with the theoretical model of the ship
- ▶ Benchmark on 2 000 000 points: foreign objects **perfectly isolated**, with no training at all



Reference model in red, anomalies in colour [AYANA RA 25]
© Naval Group

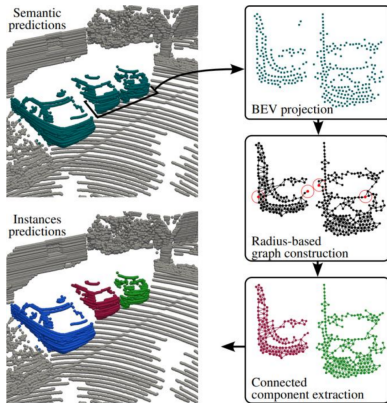
In collaboration with Marie ASPRO (Inria Startup Studio)

3D panoptic: the state of the art *is* clustering

Panoptic = the *class* of each point + its *instance*

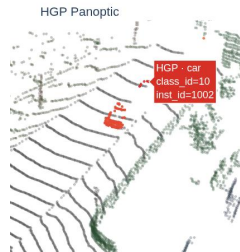
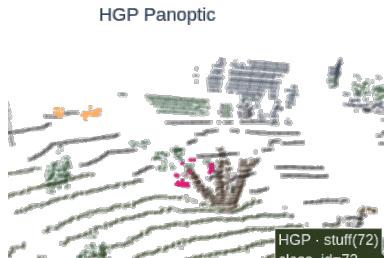
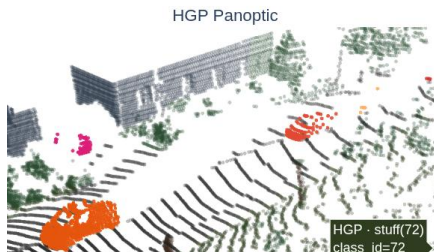
- ▶ Instances used to be learned end to end, with costly annotations
- ▶ ALPINE [ALPINE 25]: good semantics, then, class by class, **classical clustering** (k -nearest neighbour graph, threshold, **connected components**)
- ▶ In 4D, Geo-4D [Geo-4D 25] keeps the same pipeline and adds a **geometric association** between frames, also training-free

State of the art in panoptic segmentation on outdoor LiDAR



[ALPINE 25] C. Sautier, G. Puy, A. Boulch, R. Marlet, V. Lepetit, "Is clustering enough for LiDAR instance segmentation? A state-of-the-art training-free baseline" (ALPINE), 2025.
[Geo-4D 25] G. Oh, Y. Jang, J. Choi, S.-J. Kang, G. Lin, S. Kim, "Training-free global geometric association for 4D LiDAR panoptic segmentation" (Geo-4D), 2025.

Our first results: 3D

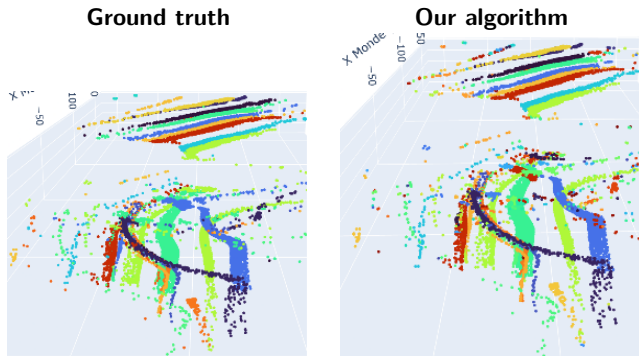


Geometric priors applied to instance volumes, on SemanticKITTI [SemanticKITTI 19]

[SemanticKITTI 19] J. Behley *et al.*, "SemanticKITTI: A Dataset for Semantic Scene Understanding of LiDAR Sequences", ICCV, 2019.



Our first results: 4D



SemanticKITTI [SemanticKITTI 19], 200 frames
tracking by **optimal transport** + KALMAN filter, **training-free**

[SemanticKITTI 19] J. Behley *et al.*, "SemanticKITTI: A Dataset for Semantic Scene Understanding of LiDAR Sequences", *ICCV*, 2019.

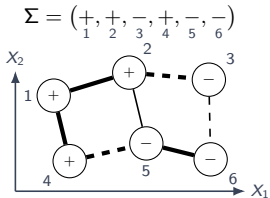
06

Towards more structured data



The two ideas of the thesis applied to more structured data

- 1) higher-order interactions bring robustness
- 2) percolation governs performance

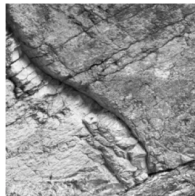


solid line: attraction dashed line: repulsion

thick line: satisfied edge (thin line: unsatisfied)

Signed graphs: the links *create* the communities

[Asilomar 25]



Images: extracting crack networks [EUVIP 26]

© Cerema

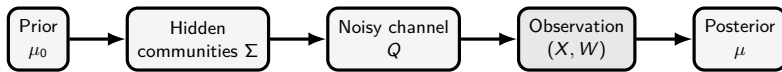
[Asilomar 25] L. Hauseux, N. Soprano-Loto, K. Avrachenkov, "Higher-order Monte Carlo cluster dynamics for community detection in Euclidean graphs", Asilomar, 2025.

[EUVIP 26] L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia, "Multi-modal, training-free crack extraction via generalized Frangi graph", EUVIP, 2026.



Community detection on signed graphs: a Bayesian framework

Community detection [Avrachenkov–Dreveton 22]: recover the hidden communities Σ of a graph whose links are *signed*



- ▶ $W_e > 0$ encodes an **attraction**; $W_e < 0$, a **repulsion**; $|W_e|$ measures the interaction strength
- ▶ In the case we study, the calibrated channel gives a signed GIBBS posterior [I&I 26+]:

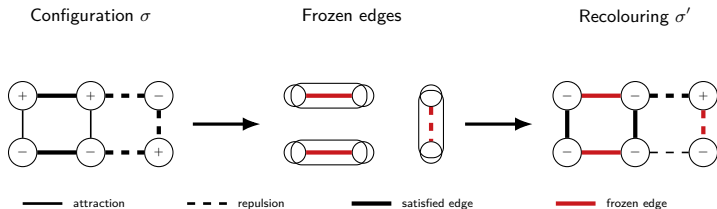
$$U(\sigma) = -\log \mu_0(\sigma \mid X) + \sum_{\substack{e=\{i,j\} \\ W_e > 0}} W_e \mathbb{I}_{\{\sigma_i \neq \sigma_j\}} + \sum_{\substack{e=\{i,j\} \\ W_e < 0}} |W_e| \mathbb{I}_{\{\sigma_i = \sigma_j\}}, \quad \mu(\sigma) = \frac{e^{-U(\sigma)}}{Z}$$

[Avrachenkov–Dreveton 22] K. Avrachenkov, M. Dreveton, *Statistical Analysis of Networks*, Now Publishers, 2022.

[I&I 26+] L. Hauseux, N. Soprano-Loto, K. Avrachenkov, "A Bayesian framework for community detection on signed graphs", paper in preparation for *Information and Inference*, 2026.



A signed Swendsen–Wang step



$$\mathbb{P}(e \text{ frozen} \mid \sigma) = \begin{cases} 1 - e^{-|W_e|}, & e \text{ satisfied} \\ 0, & e \text{ unsatisfied} \end{cases}$$

- For two classes and a uniform prior, each frozen component is **flipped as a whole** with probability 1/2: the transition is reversible and preserves μ [Swendsen–Wang 87]
- In the proof, one step from Σ produces a **coupled replica of the posterior** $\Sigma^{(1)}$

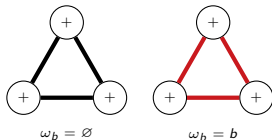
[Swendsen–Wang 87] R. H. Swendsen, J.-S. Wang, "Nonuniversal critical dynamics in Monte Carlo simulations", *Physical Review Letters*, 1987.



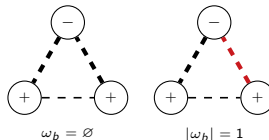
Generalising Swendsen–Wang to higher-order interactions

$$U(\sigma) = U_0(\sigma) + \sum_{b \in \mathcal{B}} U_b(\sigma), \quad b = \text{edge, triangle, clique...}$$

Attractive triangle
all or nothing



Repulsive triangle (frustrated)
never both edges

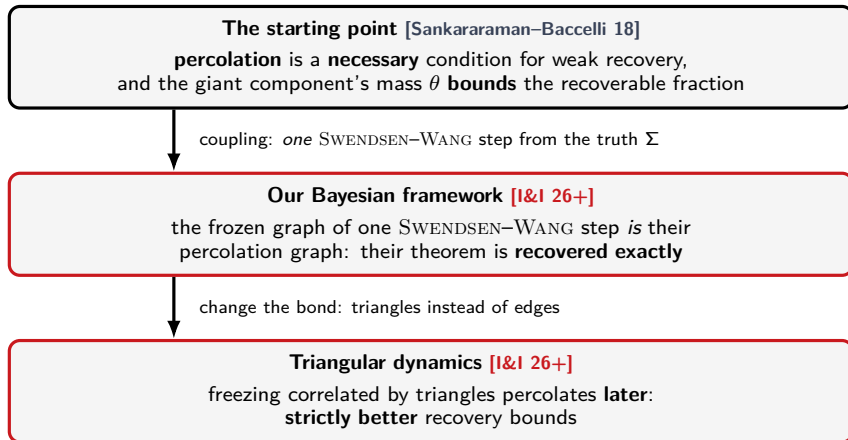


► A single draw per bond: this introduces **correlation** into the freezing process **[Asilomar 25] [I&I 26+]**

[Asilomar 25] L. Hauseux, N. Soprano-Loto, K. Avrachenkov, “Higher-order Monte Carlo cluster dynamics for community detection in Euclidean graphs”, Asilomar, 2025.

[I&I 26+] L. Hauseux, N. Soprano-Loto, K. Avrachenkov, “A Bayesian framework for community detection on signed graphs”, paper in preparation for *Information and Inference*, 2026.

Percolation bounds the recovery of communities

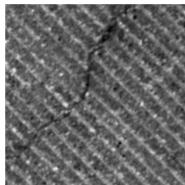


[Sankararaman–Baccelli 18] A. Sankararaman, F. Baccelli, "Community detection on Euclidean random graphs", *ACM–SIAM SODA*, 2018.

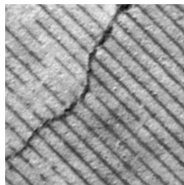
[I&I 26+] L. Hauseux, N. Soprano-Loto, K. Avrachenkov, "A Bayesian framework for community detection on signed graphs", paper in preparation for *Information and Inference*, 2026.



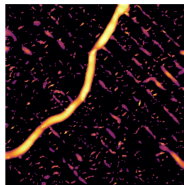
Extracting crack networks from images



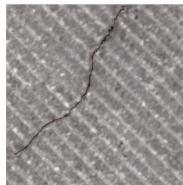
intensity



depth



edge similarity



extracted skeleton

- From the **pixel** (FRANGI filter [Frangi 98], which judges each pixel alone and fragments under noise) to the **pixel pair** (FRANGI graph [GRETSI 25])
- **Multimodal** at the **operator** level: we sum the normalised Hessians, never the responses [EUVIP 26]
- Exploratory transfers without retraining; output as an **editable graph**

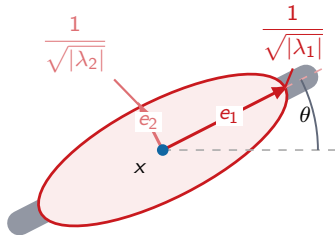
[Frangi 98] A. F. Frangi, W. J. Niessen, K. L. Vincken, M. A. Viergever, "Multiscale vessel enhancement filtering", MICCAI, 1998.

[GRETSI 25] L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia, "Généralisation du filtre de Frangi pour l'extraction des réseaux de fissures", GRETSI, 2025.

[EUVIP 26] L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia, "Multi-modal, training-free crack extraction via generalized Frangi graph", EUVIP, 2026.

The Hessian: information about *tubular* structures

$\mathcal{H}_\sigma(x) = \nabla^2(G_\sigma * I)(x)$, real symmetric: principal axes e_1, e_2 , semi-axes $1/\sqrt{|\lambda_1|}$ and $1/\sqrt{|\lambda_2|}$



The FRANGI filter [Frangi 98] draws **two** numbers from it:

- ▶ $R_B = \frac{|\lambda_1|}{|\lambda_2|}$: the **shape**
0 for a very elongated ridge, 1 for a disk
- ▶ $S = \sqrt{\lambda_1^2 + \lambda_2^2}$: the **contrast**
the strength of the curvatures

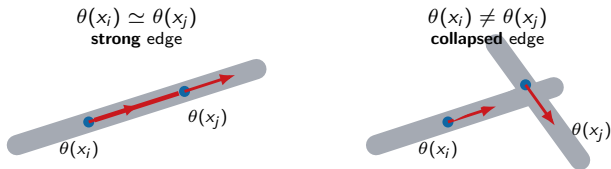
$$\exp\left(-\frac{R_B^2}{2\beta^2}\right) \left(1 - \exp\left(-\frac{S^2}{2c^2}\right)\right) \quad \text{if } \lambda_2 > 0, \quad 0 \text{ otherwise}$$

the FRANGI response: a dark, high-contrast ridge



What the graph adds: *alignment*

The filter judges each pixel **alone**; the graph requires the directions e_1 of the **two** pixels to agree

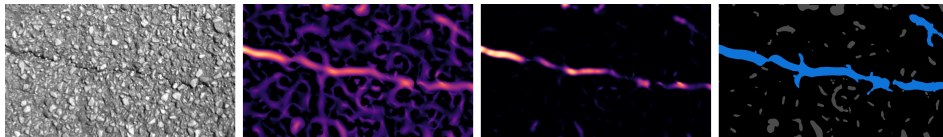


$$S_{\text{align}} = \exp \left[-\frac{1}{2} \left(\frac{\sin(\theta(x_i) - \theta(x_j))}{s_a} \right)^2 \right]$$

the sine, because θ and $\theta + \pi$ describe the same axis [GRETSI 25] [EUVIP 26]

[GRETSI 25] L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia, "Généralisation du filtre de Frangi pour l'extraction des réseaux de fissures", GRETSI, 2025.
[EUVIP 26] L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia, "Multi-modal, training-free crack extraction via generalized Frangi graph", EUVIP, 2026.

A granular background: order 2 cuts accidental bridges



granular visible image

pixel-wise FRANGI response

FRANGI similarity

component kept in blue

- ▶ Every pebble carries a response: the background is as much a “ridge” as the crack
- ▶ A **chain of edges** is then enough to link the pebbles one to the next: the background percolates
- ▶ At **order 2**, two edges are glued only through a **triangle**: a chain with no triangle no longer propagates connectivity, and the percolation rate increases with K **[ANS 26]**: hence the choice $K = 2$ **[EUVIP 26]**

[ANS 26] L. Hauseux, K. Avrachenkov, J. Zerubia, “Generalization of Single-Linkage with higher-order interactions”, *Applied Network Science*, 11(9), 2026.

[EUVIP 26] L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia, “Multi-modal, training-free crack extraction via generalized Frangi graph”, EUVIP, 2026.

07

Perspectives: the **hierarchical** path

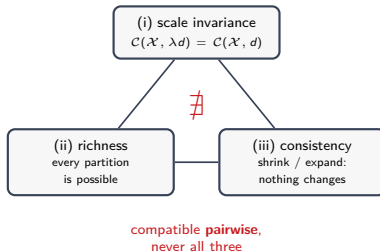


Back to the start: Kleinberg's impossibility theorem

A clustering function $\mathcal{C}$ assigns to a finite metric space $(\mathcal{X}, d)$ a partition of $\mathcal{X}$

- (i) **Scale invariance:** $\mathcal{C}(\mathcal{X}, \lambda d) = \mathcal{C}(\mathcal{X}, d)$
- (ii) **Richness:** every partition is possible
- (iii) **Consistency:** shrinking a cluster returned by $\mathcal{C}$, or pulling two of them apart, changes nothing

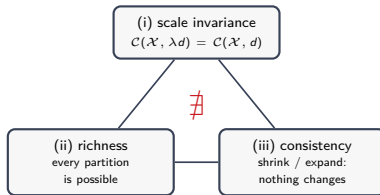
Impossibility [Kleinberg 02]: no $\mathcal{C}$ satisfies all three



Back to the start: Kleinberg's impossibility theorem

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compatible pairwise,
never all three

Impossibility [Kleinberg 02]: no $\mathcal{C}$ satisfies all three

CARLSSON & MÉMOLI found the solution, and it is **hierarchical**: it is **Single-Linkage**
[Carlsson-Mémoli 10]

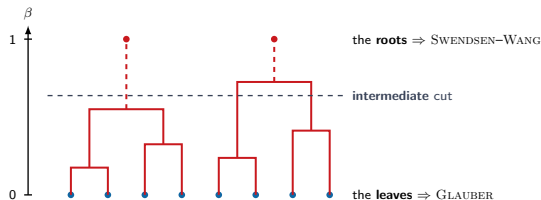
[Kleinberg 02] J. Kleinberg, "An impossibility theorem for clustering", *Advances in Neural Information Processing Systems* (NIPS), vol. 15, MIT Press, p. 446–453, 2002.
[Carlsson-Mémoli 10] G. Carlsson, F. Mémoli, "Characterization, stability and convergence of hierarchical clustering methods", *JMLR*, 2010.



A hierarchical cluster dynamics

$\xi_e \sim \text{Exp}(|W_e|)$ if e is satisfied, $\xi_e = +\infty$ otherwise;

$\Pi_\beta =$ connected components of the graph $(V, \{e : \xi_e \leq \beta\})$, $0 \leq \beta \leq 1$

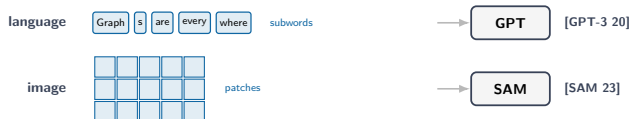


- ▶ Cutting at the **leaves** gives GLAUBER; cutting at the **roots**, SWENDSEN-WANG
- ▶ The most promising cut for **weak recovery**: the **percolation temperature** [I&I 26+]

[I&I 26+] L. Hauseux, N. Soprano-Loto, K. Avrachenkov, "A Bayesian framework for community detection on signed graphs", paper in preparation for *Information and Inference*, 2026.



Foundation models: text, and now images



- ▶ Trained **only once**, on massive data and **without annotation** [Bommasani 21]
- ▶ **Reusable**: a single base model serves many different tasks
- ▶ Capabilities **emerge** that nobody programmed
- ▶ Everyone starts from the **same** base model

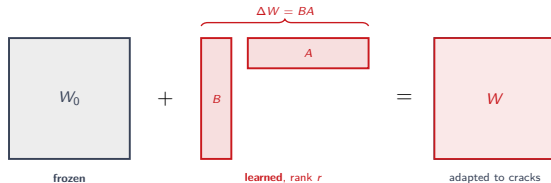
[Bommasani 21] R. Bommasani *et al.*, "On the opportunities and risks of foundation models", 2021.

[GPT-3 20] T. Brown *et al.*, "Language models are few-shot learners", *NeurIPS*, 2020.

[SAM 23] A. Kirillov *et al.*, "Segment Anything", *ICCV*, 2023.



Adapting it without retraining: *LoRA*



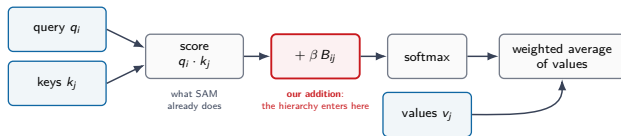
- ▶ We **freeze** the pre-trained backbone and learn only a **low-rank** update $W = W_0 + BA$ [LoRA 22]
- ▶ **CrackSAM** [CrackSAM 24]: 4.4 M trainable weights out of SAM's 637 M
- ▶ SAM has already seen everything: our extra descriptors (Hessian, gradient) change **nothing**
- ▶ And it **transfers**: cracks on sites never seen before, with no retraining

[LoRA 22] E. J. Hu *et al.*, "LoRA: low-rank adaptation of large language models", *ICLR*, 2022.

[CrackSAM 24] K. Ge *et al.*, "Fine-tuning vision foundation model for crack segmentation in civil infrastructures", *Construction and Building Materials*, 2024.



How to guide attention *via* the hierarchy? An idea to explore



- ▶ **Attention:** each token sends a **query**, compares it with the **keys** of all the others, and leaves with the average of their **values**, weighted by a **visual score** that SAM already computes
- ▶ We **add a bias** before the *softmax*: an exchange that is **encouraged**, never forced [Graphormer 21]
- ▶ The bias comes from the FRANGI graph **hierarchy**: the earlier two tokens **merge** in the tree, the stronger it is, even far apart in the image [EUVIP 26]; LoRA learns it with a **single** coefficient β [LoRA 22]

$$\text{attention} = \text{softmax}(\text{visual scores} + \beta B_H)$$

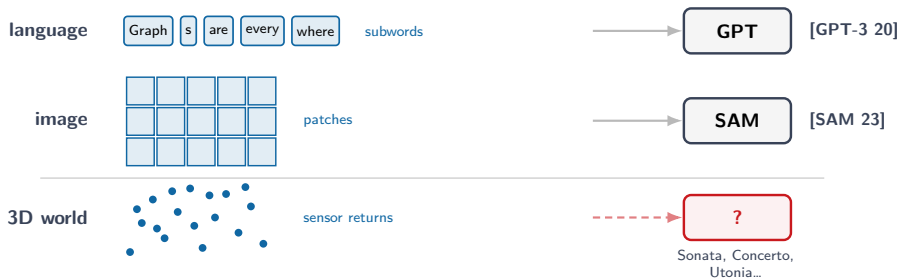
[Graphormer 21] C. Ying *et al.*, "Do Transformers really perform badly for graph representation?", *NeurIPS*, 2021.

[EUVIP 26] L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia, "Multi-modal, training-free crack extraction via generalized Frangi graph", *EUVIP*, 2026.

[LoRA 22] E. J. Hu *et al.*, "LoRA: low-rank adaptation of large language models", *ICLR*, 2022.



Towards a foundation model for 3D?

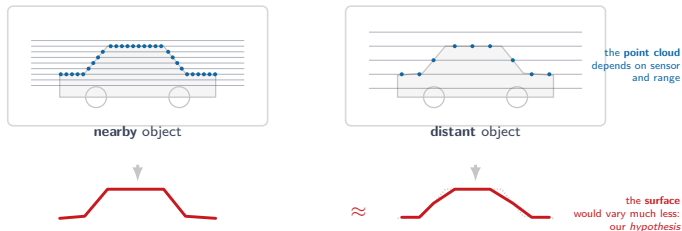


in 3D, the unit of computation is still the **sampling** unit:
sensor return or grid cell

[GPT-3 20] T. Brown *et al.*, "Language models are few-shot learners", *NeurIPS*, 2020.

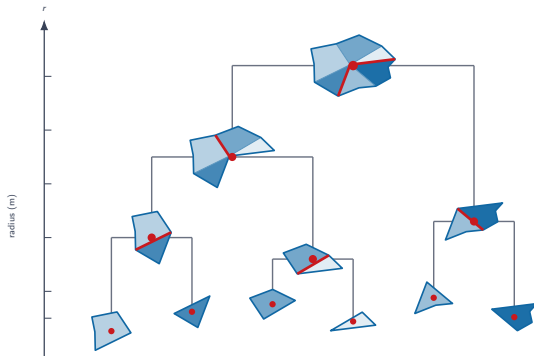
[SAM 23] A. Kirillov *et al.*, "Segment Anything", *ICCV*, 2023.

The deeper reason: the point cloud is a *sensor artefact*



- **Density** decreases with range; on top of that come the scan lines and the **occlusions**
- So a point-by-point model learns *at the same time* the geometry, the sensor, the density, the occlusion **and** the semantics
- Hypothesis: this is *one* central reason for the difficulty

Replacing points by a hierarchy of **polyhedra**?



- Same recipe as for text: the **polyhedron** is the **alphabet** (the unit, fixed once and for all) and the **hierarchy** is its **grammar**

08

Conclusion





Two guiding threads, four contributions

1) When pairwise relations are insufficient or unstable,
higher-order interactions bring robust coherence.

2) **Percolation** is the theoretical keystone
for quantifying the performance of these algorithms

- ▶ A **unified view** of *Single-Linkage*: heuristic, geometric, statistical [ANS 26]; training-free 3D and 4D LiDAR
- ▶ **HGP-Clusterer**: the *exact* K -NN hierarchy, via Čech, GABRIEL and order- K DELAUNAY [EUSIPCO 24] [CN 24] [ANS 26]
- ▶ The **percolation rate**: it explains the failure of Robust SL and (H)DBSCAN, and the gains of K -polyhedra [SIGMETRICS SRC 24]
- ▶ **Exporting the paradigm**: signed graphs [Asilomar 25] [I&I 26+]; imaging, multimodal FRANGI graph [GRETSI 25] [EUVIP 26] [ISPRS 26+]

[EUSIPCO 24] L. Hauseux, K. Avrachenkov, J. Zerubia, "Benefits of hypergraphs for density-based clustering", *EUSIPCO*, Lyon, 2024.

[CN 24] L. Hauseux, K. Avrachenkov, J. Zerubia, "Hypergraphs, percolation and hierarchical clustering", *Complex Networks & Their Applications*, Istanbul, 2024.

[ANS 26] L. Hauseux, K. Avrachenkov, J. Zerubia, "Generalization of Single-Linkage with higher-order interactions", *Applied Network Science*, 11(9), 2026.

[SIGMETRICS SRC 24] L. Hauseux, "How can we theoretically measure the performance of density-based clustering algorithms?", ACM SIGMETRICS SRC, Venice, 2024 (second prize).

[Asilomar 25] L. Hauseux, N. Soprano-Loto, K. Avrachenkov, "Higher-order Monte Carlo cluster dynamics for community detection in Euclidean graphs", *Asilomar*, 2025.

[I&I 26+] L. Hauseux, N. Soprano-Loto, K. Avrachenkov, "A Bayesian framework for community detection on signed graphs", paper in preparation for *Information and Inference*, 2026.

[GRETSI 25] L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia, "Généralisation du filtre de Frangi pour l'extraction des réseaux de fissures", *GRETSI*, 2025.

[EUVIP 26] L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia, "Multi-modal, training-free crack extraction via generalized Frangi graph", *EUVIP*, 2026.

[ISPRS 26+] L. Hauseux, R. Antoine, P. Foucher, P. Charbonnier, J. Zerubia, paper in preparation for the *ISPRS Journal of Photogrammetry and Remote Sensing*, 2026.



Further perspectives

- ▶ **High dimension:** a higher-order **dimensionality reduction**, where current methods rely on neighbourhood graphs (UMAP)
- ▶ **Open question:** HGP-Clusterer is only *fractionally* consistent; a consistent algorithm in dimension $p \geq 2$ remains to be invented

Publications

Journal

- ▶ **[ANS 26]** *Generalization of Single-Linkage with higher-order interactions*, *Applied Network Science*, 2026

International and national conferences

- ▶ **[CN 23]** Cosmic filaments, Complex Networks, Menton, 2023
- ▶ **[EUSIPCO 24]** Hypergraphs for density-based *clustering*, EUSIPCO, Lyon, 2024
- ▶ **[CN 24]** Hypergraphs, percolation and hierarchical *clustering*, Complex Networks, Istanbul, 2024
- ▶ **[SIGMETRICS SRC 24]** Theoretical performance measure, ACM SIGMETRICS SRC, Venice, 2024 (**second prize**)
- ▶ **[Asilomar 25]** Higher-order cluster dynamics, Asilomar, 2025
- ▶ **[GRETSI 25]** Generalisation of the Frangi filter, GRETSI, Strasbourg, 2025
- ▶ **[EUVIP 26]** Training-free multimodal crack extraction, EUVIP, Luxembourg, 2026

In progress

- ▶ **[I&I 26+]** Bayesian framework for community detection on signed graphs (journal)
- ▶ **[ISPRS 26+]** Generalised, multimodal and training-free Frangi graph (ISPRS journal)

Software (*open source*): GitHub repositories Ayana-Inria/HypergraphPercol_K2 and Ayana-Inria/Frangi-EUVIP

Thank you.





Author's work presented today

- ▶ **[HAL 17]** L. Hauseux (supervised by B. Błaszczyszyn), "Un classificateur non supervisé utilisant les complexes simpliciaux", Inria Paris, 2017. HAL: hal-01597846.
- ▶ **[CN 23]** L. Hauseux, K. Avrachenkov, J. Zerubia, "Graph based approach for galaxy filament extraction", *Complex Networks & Their Applications*, Menton, 2023.
- ▶ **[EUSIPCO 24]** L. Hauseux, K. Avrachenkov, J. Zerubia, "Benefits of hypergraphs for density-based clustering", *EUSIPCO*, Lyon, 2024.
- ▶ **[CN 24]** L. Hauseux, K. Avrachenkov, J. Zerubia, "Hypergraphs, percolation and hierarchical clustering", *Complex Networks & Their Applications*, Istanbul, 2024.
- ▶ **[SIGMETRICS SRC 24]** L. Hauseux, "How can we theoretically measure the performance of density-based clustering algorithms?", ACM SIGMETRICS SRC, Venice, 2024 (second prize).
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